

GeneMeow GFP Design Concept and Documentation

Final submission based on DMS-positive brightness/supercharge axis and ColabFold validation

Team: GeneMeow

Summary. This final package uses six sfGFP-derived sequences from the DMS-positive/supercharge design axis. The final candidates were selected because they passed competition format checks, showed positive sfGFP-aware brightness delta scores, preserved the chromophore-forming TYG motif, and passed ColabFold core structural validation. The workflow does not claim experimental proof before official testing; it provides a reproducible computational prioritization pipeline.

1. Objective and scoring logic

The objective is to design GFP-family sequences with high initial fluorescence and high post-heat fluorescence retention. In the official assay, each sequence is synthesized, expressed in a cell-free protein synthesis system, measured for initial brightness, heat-treated, and measured again. The combined score is proportional to post-heat brightness relative to the WT control in the same batch:

$$\text{Comprehensive score} = (\text{F}_{\text{initial}} / \text{F}_{\text{initialWT}}) \times (\text{F}_{\text{final}} / \text{F}_{\text{initial}}) = \text{F}_{\text{final}} / \text{F}_{\text{initialWT}}$$

Therefore, the design strategy must protect initial brightness while increasing the chance of retaining fluorescence after heat treatment.

2. Final computational workflow

- Load official references, GFP brightness data, exclusion list and submission template.
- Use sfGFP as the parent scaffold and preserve the chromophore-forming TYG motif at positions 65-67.
- Evaluate initial brightness using an sfGFP-aware delta model derived from avGFP DMS coefficients, with corrected dataset numbering.
- Evaluate thermal-retention potential using a stability proxy based on negative surface-charge shift and mutation risk.
- Run ColabFold on WT sfGFP and all six candidates to check preservation of the GFP beta-barrel and chromophore environment.
- Submit a portfolio of conservative, balanced and thermal/supercharge candidates.

3. Brightness validation

Brightness was not treated as a directly predicted experimental $\text{F}_{\text{initial}}$ value. Instead, each mutation was scored as a delta effect relative to sfGFP using avGFP DMS-derived coefficients. The key correction is that the DMS dataset numbering excludes the initial methionine, so protein position P maps to dataset position P - 1. The numbering check was perfect for the available DMS features.

All six final sequences have positive sfGFP-aware brightness delta scores and all mutation effects used in this validation were measured, not estimated. This is the main advantage of the final DMS-positive design axis.

Seq	Mutations	Brightness delta	Verdict	Measured	Estimated	Ref mismatch	Low support
1	D19E:R73L:H2 31Y	0.170	>= sfGFP brightness prior	3	0	0	0
2	D19E:R73L:K1 66E:N212D:H2 31Y	0.283	>= sfGFP brightness prior	5	0	0	0
3	D19E:R73L:K1	0.346	>= sfGFP	6	0	0	0

	66E:N198D:N212D:H231Y		brightness prior				
4	D19E:R73L:K101E:K156E:K166E:N198D:N212D:H231Y	0.379	>= sfGFP brightness prior	8	0	0	0
5	D19E:R73L:H231Y:Y237N	0.279	>= sfGFP brightness prior	4	0	0	0
6	K101E:K156E:K166E:N198D:N212D	0.209	>= sfGFP brightness prior	5	0	0	0

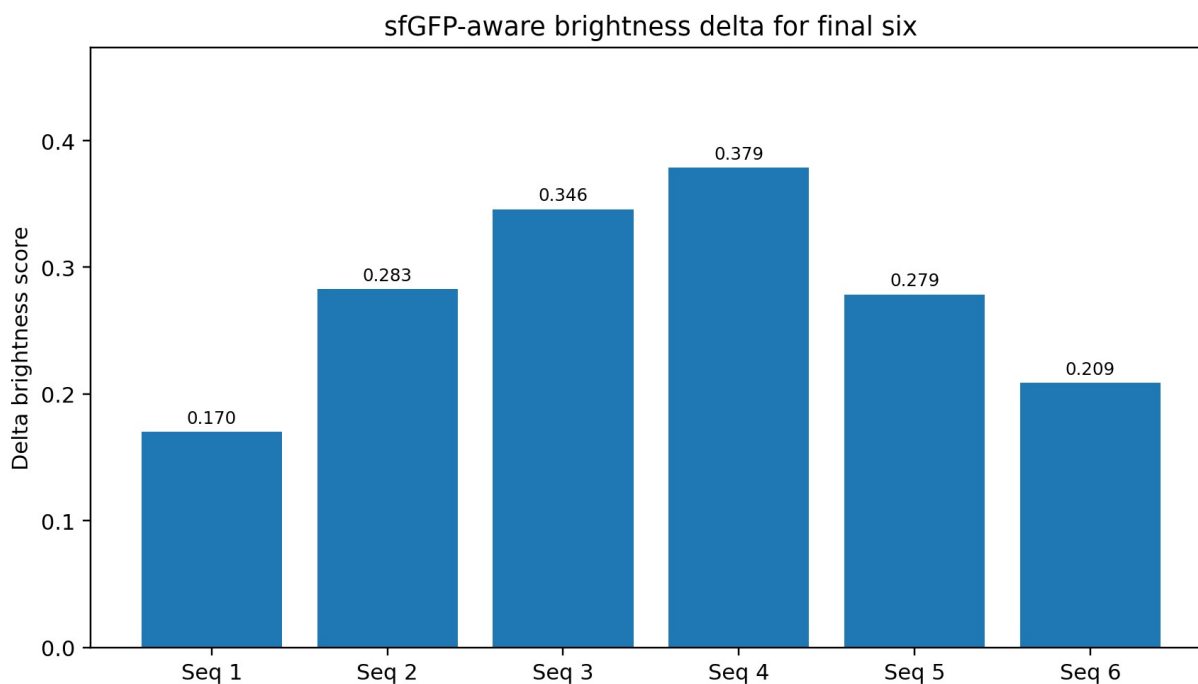


Figure 1. All six final sequences have positive sfGFP-aware brightness delta scores.

4. Stability proxy and risk control

Because no large public dataset exactly matches the official 72C heat-treatment assay, thermal retention was not predicted directly. The workflow used a stability proxy: moderate negative surface-charge shift was treated as a solubility/aggregation-resistance prior, while excessive mutational load was penalized.

The final six include conservative brightness-preserving candidates, balanced charge-shift candidates and stronger supercharge candidates. Seq 3 and Seq 6 are especially strong balanced/thermal candidates by this proxy.

Seq	Charge delta	Stability proxy	Risk penalty
1	1.10	0.316	0.240
2	4.10	1.22	0.400
3	5.10	1.51	0.480
4	9.10	1.30	1.14
5	1.10	0.316	0.420
6	8.00	1.56	0.400

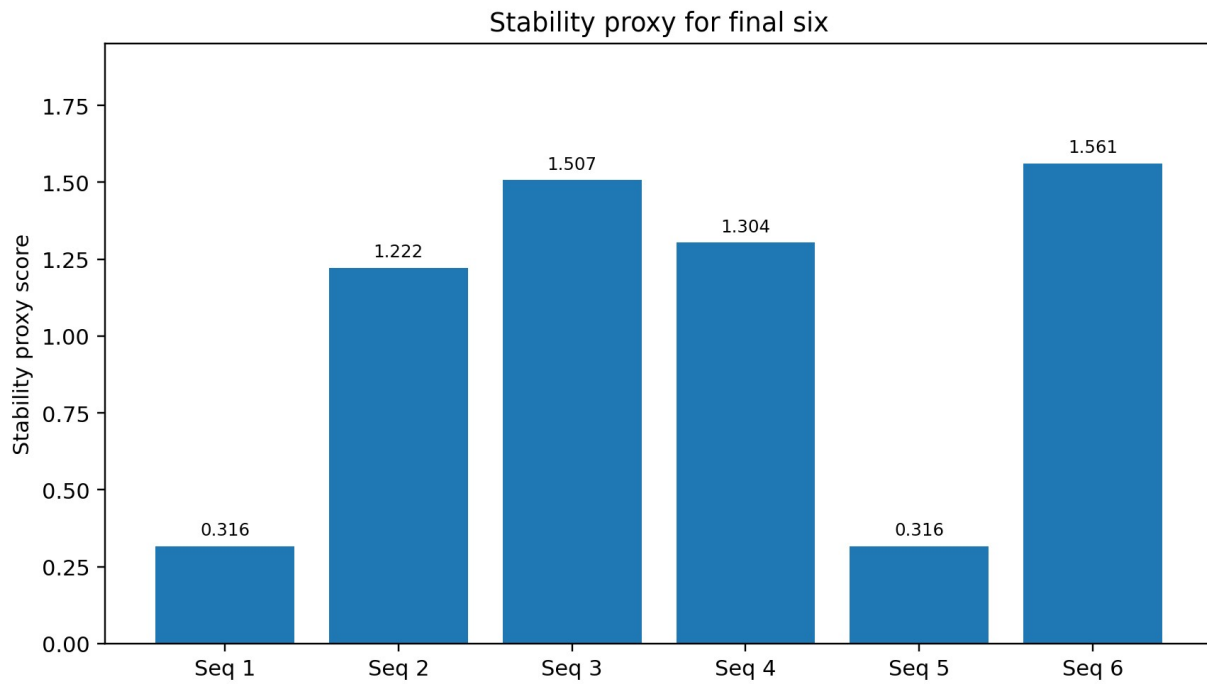


Figure 2. Stability proxy values for the final six candidates.

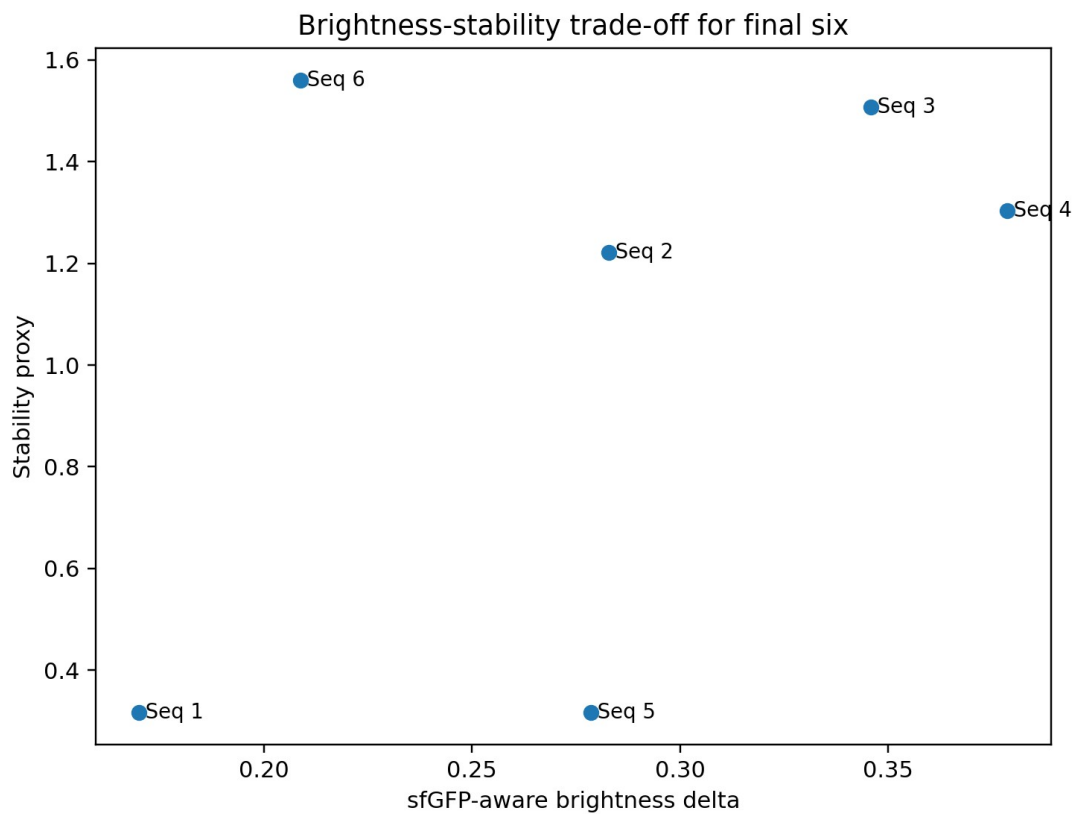


Figure 3. Brightness-stability trade-off across the final six candidates.

5. ColabFold structural validation

WT sfGFP and the six designed variants were evaluated with ColabFold AlphaFold2_batch using MMseqs2 UniRef + Environmental MSAs, five models per sequence, three recycles, no templates and no Amber relaxation. ColabFold

was used only as a structural sanity check; it does not predict fluorescence intensity, chromophore maturation or experimental heat retention.

All final candidates preserved a high-confidence GFP-like fold. Mean pLDDT values were about 96, pTM was 0.91 for all variants, chromophore-region confidence was high, and core RMSD to WT sfGFP remained below 0.13 Å. Some terminal mutations such as H231Y or Y237N have lower per-position pLDDT because the C-terminus is flexible; the core mutation-site pLDDT remains high in all variants.

Seq	Mean pLDDT	pTM	Chromo min	Core RMSD Å	Core mut-site min	Pass
1	96.21	0.910	95.38	0.125	98.12	True
2	96.17	0.910	95.75	0.040	97.75	True
3	96.23	0.910	96.00	0.042	97.12	True
4	96.33	0.910	95.31	0.123	95.88	True
5	96.33	0.910	94.88	0.128	98.12	True
6	96.22	0.910	94.75	0.115	95.75	True

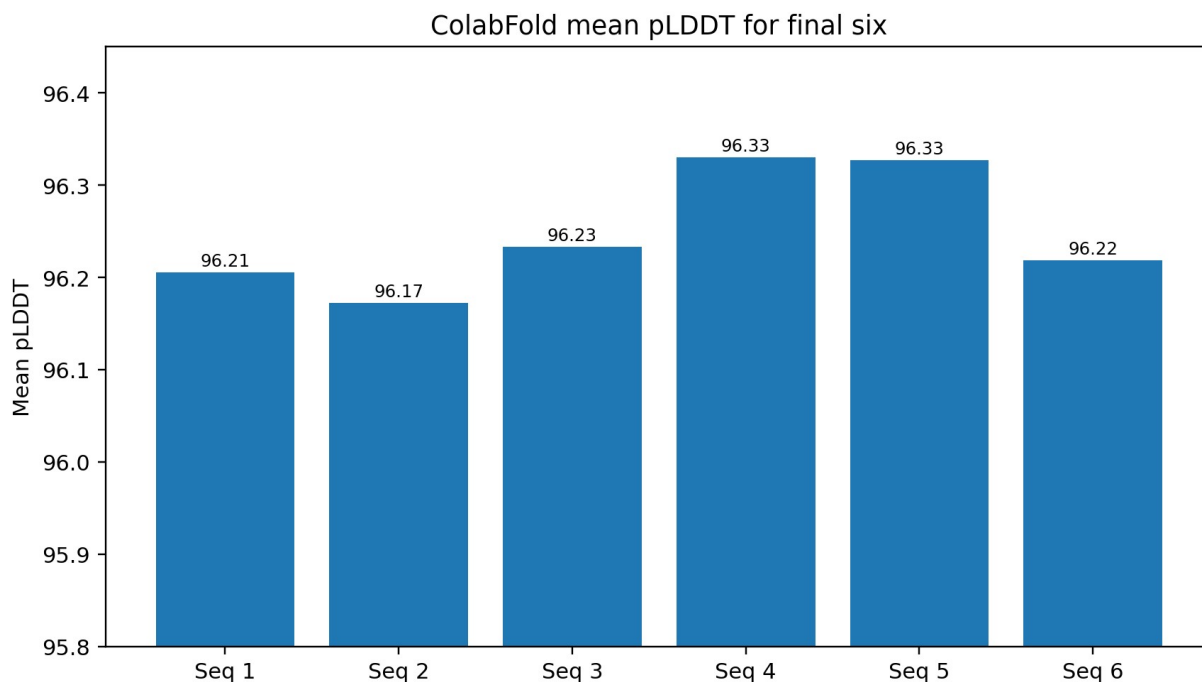


Figure 4. Mean pLDDT for the final six candidates.

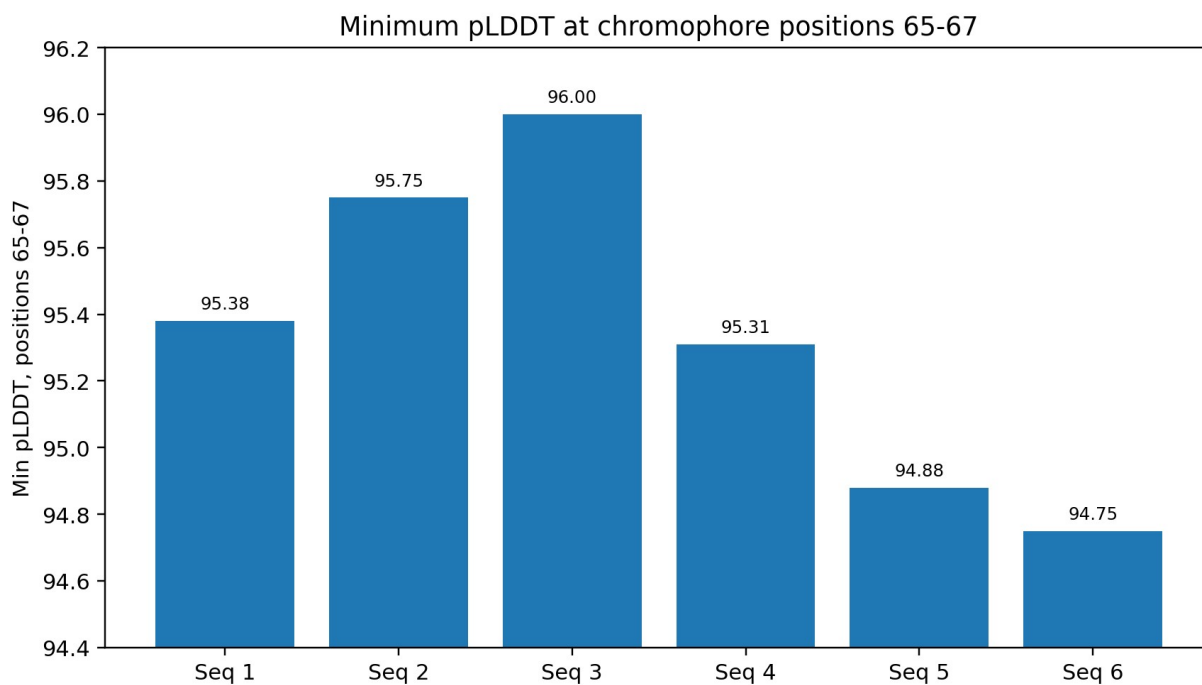


Figure 5. Minimum pLDDT in chromophore-forming positions 65-67.

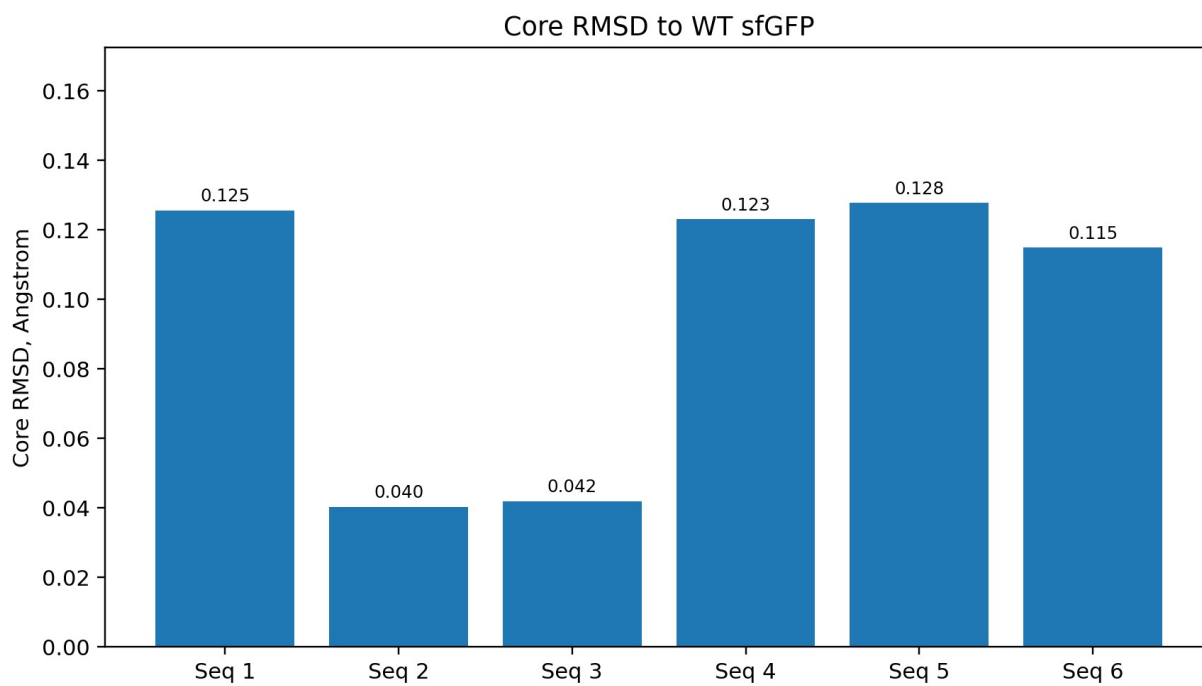
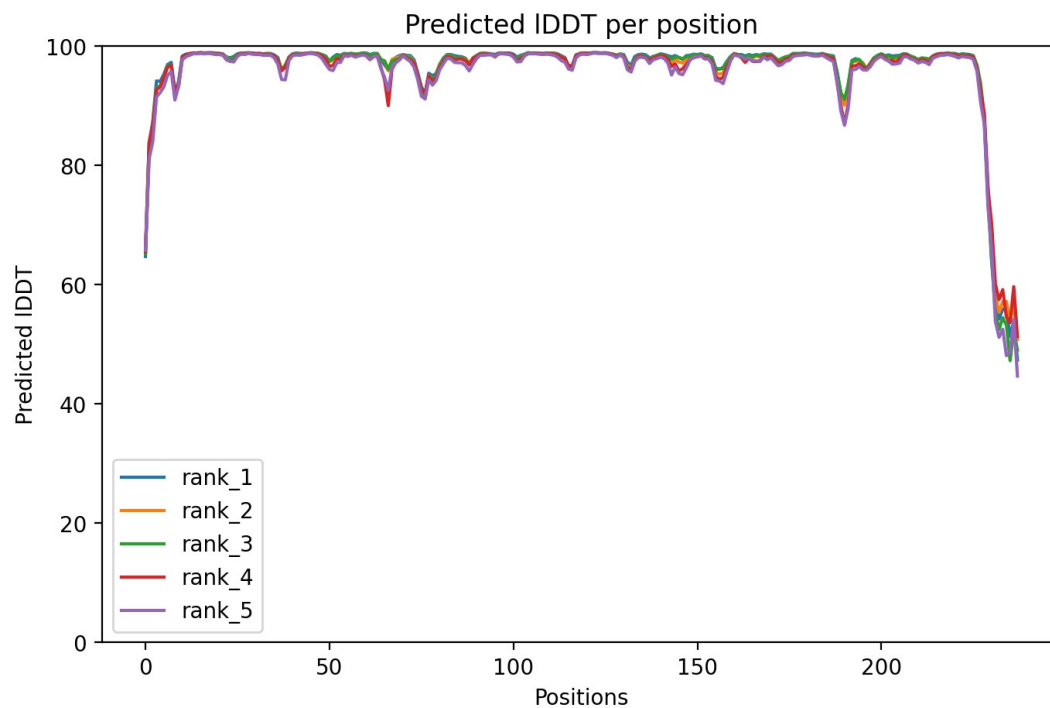


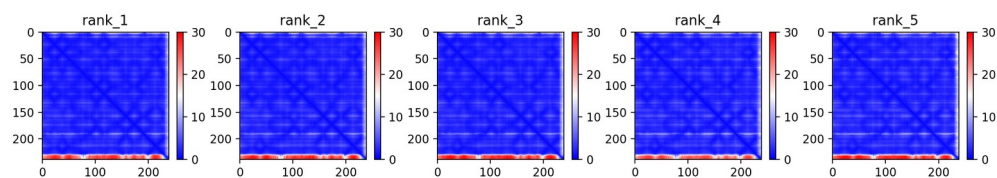
Figure 6. Core RMSD to WT sfGFP.

6. Representative ColabFold output plots

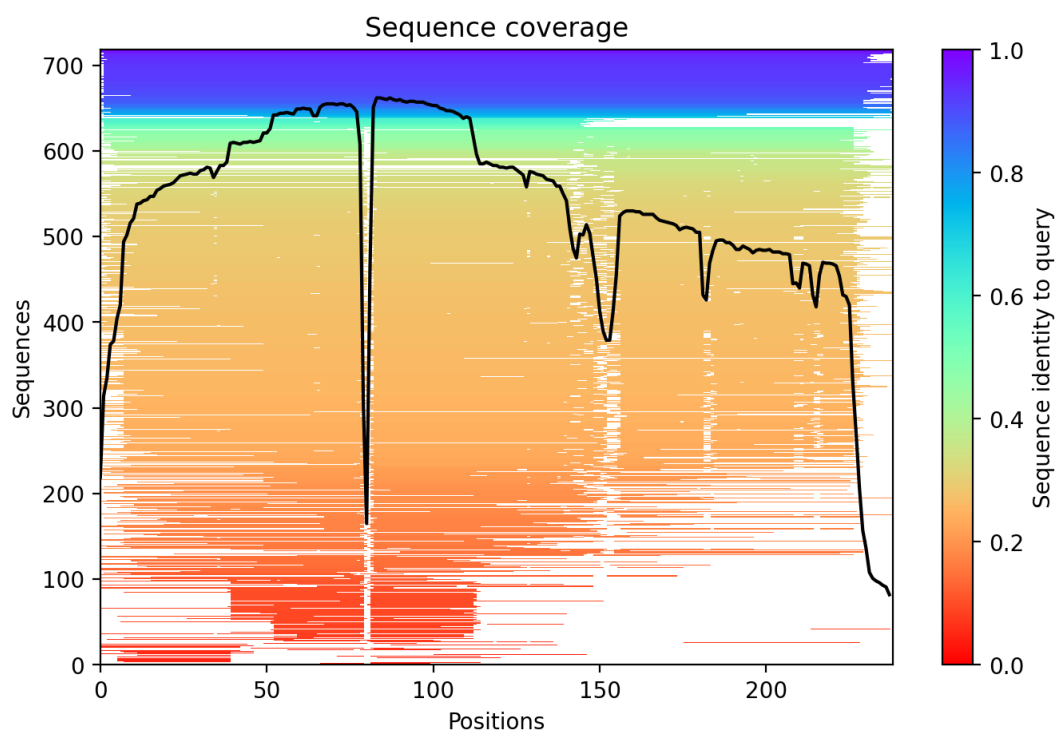
Representative raw ColabFold plots are included for the strongest balanced candidate Seq 3 and the strongest thermal/supercharge candidate Seq 6. The complete raw plots are stored in the GitHub repository.



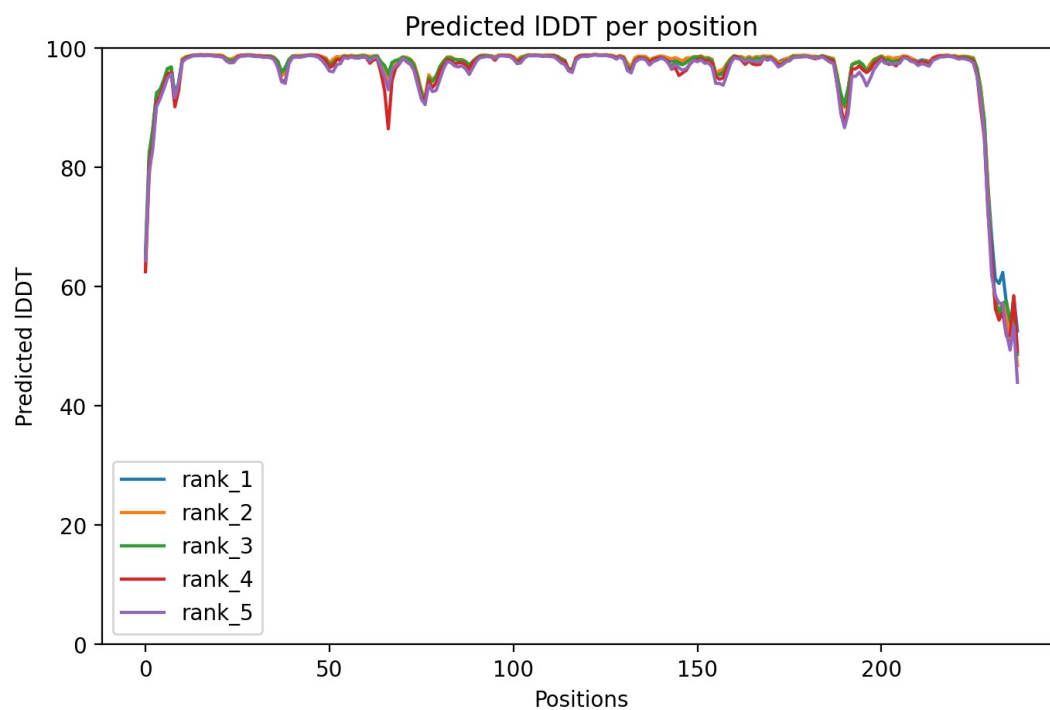
Seq 3 balanced candidate: ColabFold pLDDT plot.



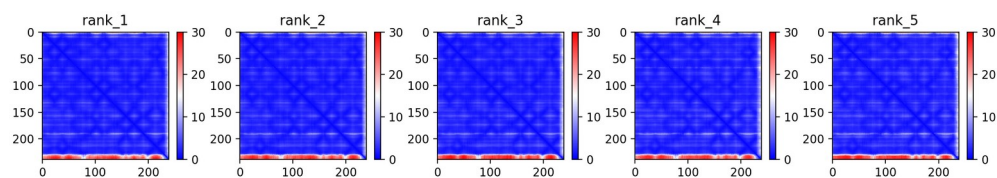
Seq 3 balanced candidate: ColabFold PAE plot.



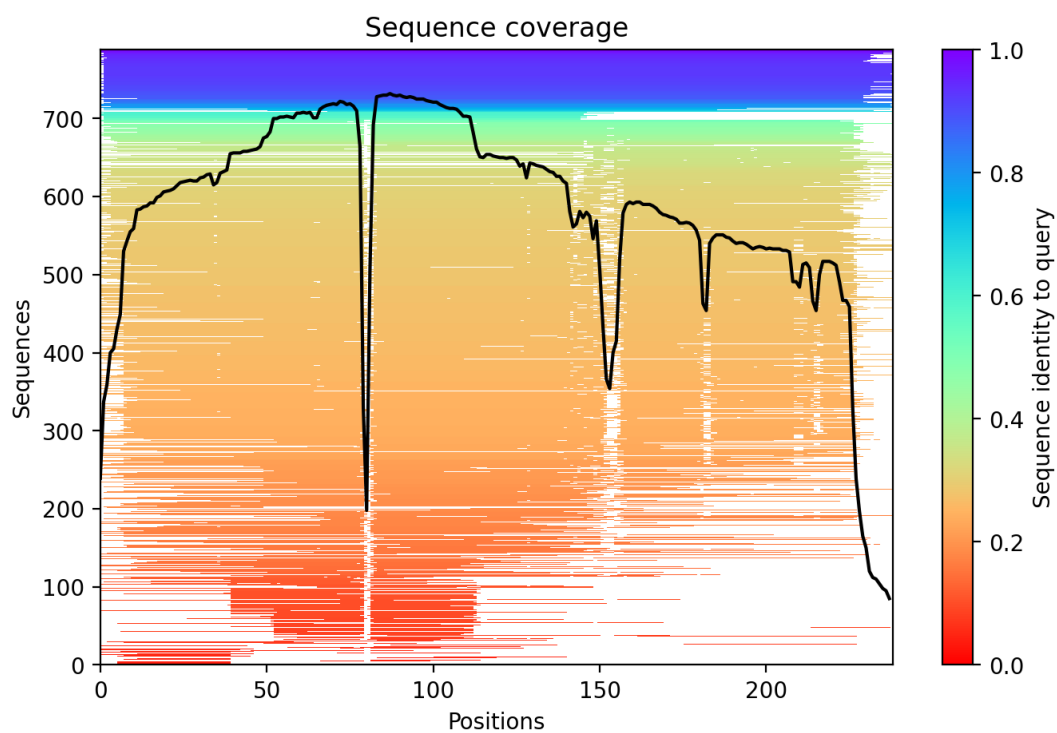
Seq 3 balanced candidate: MSA coverage plot.



Seq 6 supercharge candidate: ColabFold pLDDT plot.



Seq 6 supercharge candidate: ColabFold PAE plot.



Seq 6 supercharge candidate: MSA coverage plot.

7. Final submitted sequences

The submitted CSV contains exactly six rows with the required columns Team_Name, Seq_ID and Sequence. All sequences are 238 aa long, start with M, use only standard amino acids and preserve TYG at positions 65-67.

Seq	Role	Mutations
1	conservative surface brightness	D19E:R73L:H231Y
2	balanced core	D19E:R73L:K166E:N212D:H231Y
3	main balanced	D19E:R73L:K166E:N198D:N212D:H231Y
4	thermal high-risk/high-reward	D19E:R73L:K101E:K156E:K166E:N198D:N212D:H231Y
5	brightness backup	D19E:R73L:H231Y:Y237N
6	supercharge insurance	K101E:K156E:K166E:N198D:N212D

8. Reproducibility package

The accompanying GitHub-ready repository includes the final submission CSV, validation metrics, ColabFold plots, rank-001 PDB/JSON files, scripts for brightness scoring and ColabFold result processing, and this documentation.

9. Limitations

- Brightness delta is a DMS-derived computational prior, not a direct experimental F_{initial} measurement.
- The stability proxy is not a melting temperature and does not directly predict F_{final} after 72C heat treatment.
- ColabFold validates structural plausibility but does not model chromophore maturation, quantum yield, extinction coefficient, or CFPS expression yield.
- Final performance must be determined by the official experimental assay.

10. LLM Agent usage, logic tree and key execution logs

Large language models and code agents were used as interactive assistants for reasoning, code generation, validation, documentation and artifact packaging. The team retained responsibility for domain decisions, final sequence selection and correction of model outputs. This section provides the compact logic tree and key execution logs required for reproducibility; the complete combined log is included in the GitHub repository as docs/agent_workflow_combined_GeneMeow.md.

10.1 Agent roles and human-in-the-loop controls

- Human team: defined the biological objective, reviewed candidate designs, challenged weak assumptions and approved the final six sequences.
- LLM/code agents: generated and revised scripts, compared design branches, summarized evidence, created validation tables and assembled submission artifacts.
- Quality control: every candidate had to pass format checks, exclusion-list screening, sfGFP-aware brightness validation and ColabFold structural sanity checks before being considered for final submission.

10.2 Logic tree

```
GOAL: select six GFP-family sequences for high post-heat brightness
|
+-- Load official competition materials
|   +-- reference GFP sequences, GFP_data.xlsx, Exclusion_List.csv, submission template
|
+-- Build initial brightness evidence
|   +-- train Ridge + isotonic model on avGFP brightness DMS data
|   +-- verify DMS numbering offset: dataset_position = protein_position - 1
|   +-- score mutations as delta effects relative to sfGFP, not as absolute avGFP-to-sfGFP predictions
|
+-- Build stability evidence
|   +-- use sfGFP as robust scaffold
|   +-- add negative surface-charge / supercharging mutations
|   +-- penalize excessive mutational load and risky designs
|
+-- Generate and audit candidate designs
|   +-- check sequence length, alphabet, initial M and TYG chromophore preservation
|   +-- remove exact matches to Exclusion_List.csv and previously listed sequences
|   +-- compare H148S-heavy branch and DMS-positive/supercharge branch
|
+-- Structural validation
|   +-- run ColabFold AlphaFold2_batch on WT sfGFP and all final candidates
|   +-- extract mean pLDDT, pTM, chromophore pLDDT and core RMSD to WT
|
+-- Final selection and packaging
|   +-- choose final six with positive brightness delta and structural pass
|   +-- create submission_GeneMeow.csv, design PDF and GitHub repository
```

10.3 Key execution logs

```
[1] Reference and format checks
sfGFP length = 238 aa; chromophore positions 65-67 = TYG
Final CSV rows = 6; all sequences start with M; all contain only 20 standard amino acids
Exclusion_List.csv check = no exact matches

[2] Brightness model / corrected sfGFP-aware scoring
avGFP brightness rows used = 51715
mutation features = 1703
raw Ridge test R2 = 0.6935
Ridge + isotonic test R2 = 0.9214; MAE = 0.1700
numbering offset check = 1703/1703 tokens match avGFP at protein_position - 1
final scoring mode = mutation-level delta relative to sfGFP

[3] Final six brightness validation
Seq 1 delta = +0.1700
Seq 2 delta = +0.2829
Seq 3 delta = +0.3459
Seq 4 delta = +0.3787
Seq 5 delta = +0.2786
Seq 6 delta = +0.2087
result = all six have positive sfGFP-aware brightness delta scores
```

```
[4] ColabFold validation
  number of final sequences evaluated = 6 plus WT sfGFP
  mean pLDDT range = 96.17-96.33
  pTM = 0.91 for all final candidates
  chromophore pLDDT minimum range = 94.75-96.00
  core RMSD to WT sfGFP range = 0.040-0.128 Å
  result = all final sequences passed structural sanity checks
```

10.4 Self-correction and audit trail

The design process included an explicit self-falsification step. An earlier H148S-heavy GeneMeow branch passed ColabFold but showed weaker brightness-delta evidence. The final DMS-positive/supercharge branch was selected after it showed positive sfGFP-aware brightness delta scores for all six variants while also passing ColabFold structural checks.

Audit step	Observed issue	Correction	Effect on final design
Full-sequence avGFP to sfGFP prediction	Predictions saturated and did not distinguish sfGFP variants	Use mutation-level delta scoring relative to sfGFP	Avoided false absolute brightness claims
H148S-heavy branch audit	Several candidates had negative brightness-delta evidence	Compared against DMS-positive/supercharge branch	Final six changed to the DMS-positive/supercharge set
ColabFold structural audit	C-terminal mutations H231Y/Y237N had lower local pLDDT in flexible tail	Evaluate core mutation-site pLDDT separately from tail pLDDT	No beta-barrel or chromophore disruption detected
Submission package check	Potential risk of bloated archives from full ColabFold result extraction	Use memory-safe packaging and include only essential outputs	Repository remains reproducible and compact

10.5 Reproducibility pointers

- Full agent workflow log: docs/agent_workflow_combined_GeneMeow.md
- Final metrics table: outputs/final6_colabfold_brightness_metrics.csv
- Submission file: outputs/submission_GeneMeow.csv
- ColabFold plots and rank-001 structures: figures/ and models/colabfold_rank001_final6/